

GENERAL
MEASURES OF BATTING PERFORMANCE IN A
SHORT SERIES OF CRICKET MATCHES

H.H. Lemmer

Department of Statistics, University of Johannesburg, P.O. Box 524, Auckland
Park, 2006, Johannesburg, South Africa
e-mail: *hoffiel@uj.ac.za*

Key words: Batting average, cricket, not-out scores.

Summary: The traditional average is not a satisfactory measure of batting performance if a batsman has many not-out scores. The purpose of this paper is, firstly, to define an estimator that will always give a reliable estimate of the average, even in the case of a batsman having a large percentage of not-out scores, and secondly also a comprehensive measure of batting performance for a short series, e.g. a test or ODI series. Strong arguments are given why the traditional average should not be used in the case of high proportions of not-out scores.

1. Introduction

The custom of calculating a batsman's average by dividing his total number of scores by the number of times he was out, often leads to unrealistic averages in a short series of matches. In the 1999 World Cup Series Lance Klusener scored 281 runs in eight innings and was out only twice. His highest score was 52 but his average was 140.5, which is ridiculous. To show how erratic the average (indicated by ave) can behave, consider an artificial set of scores containing long runs of not-out scores: 23, 22*, 34*, 12*, 4*, 18*, 45*, 9*, 16, 30, 54, 21, 15, 33*, 23*, 67*, 4*, 18*, 20*, 30*, 20*, 39, 22, 23, 43, 12, 34, 23, 24, 10, 50 and 16. In figure 1 the value of ave is given after each innings played, and also of a proposed estimator e_2 , which will be discussed

MSC2000: 62P99

in this study. A sequence of not-out scores causes the value of ave to increase rapidly, especially if this occurs early in the set of scores considered. Had the first score also been not out, ave would have started with a value of 183 after the ninth innings.

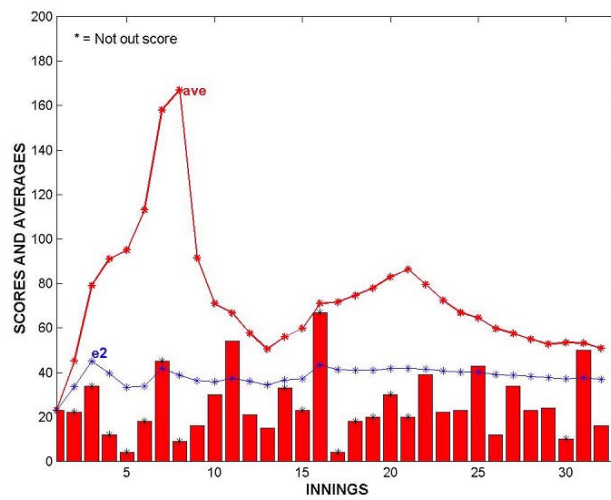


Figure 1. Different estimates of the average for artificial data

In order to compare the performances of batsmen after a test or ODI series, it is necessary to use a proper measure for his batting average. Such a measure is developed by fitting different statistical distributions to the scores of a group of batsmen and using the best of these to find a formula to calculate the averages of the batsmen. It is shown that these averages are much more reliable than the traditional cricket average.

Kimber and Hansford (1993) proposed a formula for the batting average which is based on the Kaplan-Meier estimator used in survival analysis. They mention, however, that “in cases of a huge proportion of not-out innings”, their estimator is “no more sensible than the ordinary average”. This is confirmed in the present study. In table 4 it can be seen that their measure, KH, performs almost as badly as the traditional average. A further disadvantage is that it must be recalculated from scratch when new scores arise. This estimator is obviously not a solution to the problem discussed in this paper. Buckley and James (1979) use a regression model $y_i = \alpha + \beta x_i + \varepsilon_i$, $i = 1, 2, \dots, n$ where each y_i may be censored or uncensored. For each y_i a value of the regressor x_i is required, which is, however, not available in the present problem. Their method can therefore not be used.

2. Possible estimators

Let $x_1, x_2, \dots, x_m$ denote the out scores of a batsman and $x_{m+1}^*, x_{m+2}^*, \dots, x_n^*$ the not-out scores. The traditional average, which is based on the assumption that a batsman's scores follow an exponential distribution, is given by

$$ave = \left(\sum_{i=1}^m x_i + \sum_{i=m+1}^n x_i^* \right) / m.$$

It will be shown that this formula does not give reliable values if the proportion of not-out scores becomes large. Various alternative formulas have been studied, each attempting to augment the not-out score x_i^* to a full or completed score and then dividing the sum of all the scores by n instead of m . Further discussion will focus mainly on the following type of formula:

$$e = \left(\sum_{i=1}^m x_i + f. \sum_{i=m+1}^n x_i^* \right) / n$$

where f stands for a factor which could be constant (e.g. $f = 2$) or a function of certain variables. The factor f can be seen as a constant by which the not-out scores are scaled up to obtain complete scores. Start with the former, i.e.

$$e_2 = \left(\sum_{i=1}^m x_i + f_2 \cdot \sum_{i=m+1}^n x_i^* \right) / n \text{ with } f_2 = 2.$$

The justification for the choice $f_2 = 2$ is as follows: Let X^* denote the score that the batsman will obtain if he is allowed to bat until he gets out. Due to various reasons, however, a batsman's innings can be curtailed, leaving him with a not-out score x^* . If it is assumed that external factors that can lead to a curtailment are random and independent of the batsman's potential score, use the model $x^* = X^*U$ where U is a random variable with a distribution on the interval $[0, 1]$. Then

$$\begin{aligned} E(x^*) &= E(X^*).E(U) \\ &= E(X^*).(1/2) \text{ if } U \text{ is such that } E(U) = \frac{1}{2}. \end{aligned}$$

Conversely, then, $E(X^*) = 2 E(x^*)$, which concludes the justification.

In order to find a reliable formula, it is necessary to compare the estimate it gives with the true average. To determine the true average, the distribution that fits the scores of the batsman best, has to be found. Cricket scores are discrete values and the geometric distribution is the logical model to use – cf. Barr and van den Honert (1998). It has, however, been shown in Lemmer (2001) that certain continuous distributions gave better fits to the scores than the geometric distribution or its continuous counterpart, the exponential distribution.

3. Fitting distributions to batsmen's ODI scores

The **first basic assumption** used in this study is that a batsman's scores can be modelled by means of a statistical distribution, which gives a good fit to the

aggregate of all his scores, i.e. his out scores as well as his potential complete scores in the case of not-out scores. The reasoning is that a batsman, who has a not-out score x^* , could have scored more runs, had he had the opportunity to bat until he got out. This potential or complete score X^* is calculated from the formula $X^* = E(X | X \geq x^*)$. The following distributions will be fitted:

- The exponential (λ) distribution with density function $f_X(x) = \lambda e^{-\lambda x}$, $x > 0$, $\lambda > 0$.
- The gamma (α , λ) distribution with density function $f_X(x) = \lambda^\alpha x^{\alpha-1} e^{-\lambda x} / \Gamma(\alpha)$, $x > 0$, $\lambda > 0$, $\alpha > 0$.
- The Weibull (β , λ) distribution with density function $f_X(x) = \beta \lambda^\beta x^{\beta-1} \exp(-(\lambda x)^\beta)$, $x > 0$, $\lambda > 0$, $\beta > 0$.

The parameters of a distribution are found by maximising the likelihood function over all possible values of the parameters. In the case of a not-out score x^* , the probability $P(X \geq x^*)$ is used in the likelihood function. For details of the procedure, see Lemmer (2001).

In order to develop a reliable estimator for the batting average, twenty experienced ODI batsmen who have established themselves as good batsmen, have been selected randomly from all the players currently involved in ODIs who had played in at least one hundred innings each and had averages above twenty. Vettori was later added due to his high proportion of not-out scores and Smith just for the sake of interest. (The case of test matches will be dealt with later.) For all the players concerned, their ODI scores have been obtained from Cricinfo (2005a) on 18/08/2005. The statistics of the players are given in table 1 with 'avout' the average of his out scores, 'avno' the average of his not-out scores, 'out' his number of out scores, 'no' his number of not-out scores, 'propno' his proportion of not-out scores, i.e. $\text{propno} = \text{no}/n$ and 'ave', his traditional average.

Table 1. Statistics of ODI data set

Name	<i>n</i>	avout	out	avno	no	propno	ave
Cairns	184	25.2	162	31.0	22	0.120	29.43
Kallis	206	32.3	169	60.1	37	0.180	45.41
Atapattu	223	29.7	197	60.7	26	0.117	37.74
Gibbs	168	30.1	155	64.5	13	0.077	35.53
Fleming	226	26.0	207	66.6	19	0.084	32.12
Dravid	236	33.0	207	51.1	29	0.123	40.19
Chanderpaul	166	29.7	145	46.4	21	0.127	36.42
Razaq	168	20.2	127	31.1	41	0.244	30.25
Lara	249	35.2	223	57.8	26	0.104	41.95
Gayle	120	32.9	114	111.2	6	0.050	38.74
Smith	67	37.6	64	68.3	3	0.045	40.77
Tresco hick	108	32.6	102	87.3	6	0.056	37.73
Martyn	157	23.6	109	42.4	48	0.306	42.24
Kaluwitharan	181	19.4	167	33.6	14	0.077	22.22
Boucher	125	17.8	95	21.1	30	0.240	24.47
Vettori	92	8.7	62	10.3	30	0.326	13.69
Streak	154	15.0	100	24.5	54	0.351	28.20
Klusener	137	20.6	87	35.7	50	0.366	41.10
Lehmann	101	30.1	79	31.8	22	0.218	38.96
Harris	213	17.1	151	29.0	62	0.291	29.00
Ponting	172	34.7	148	44.1	24	0.140	41.86
Pollock	145	14.8	92	17.2	53	0.366	24.67

It is interesting to notice that most batsmen have fairly high not-out averages that are often much higher than the average of their out scores. High not-out scores often appear in the case of top and middle order batsmen who had ample time on the crease to get a high score. Note, however, that under the assumptions of this study it cannot be assumed that a batsman with a high propno value will have an avno value much higher than his avout value. The **second basic assumption** in this study is that a batsman's out and not-out scores emerge basically from the same distribution. The very last ball that a batsman faces determines whether he is out or not out. Up to the penultimate ball no-one knows whether he will be out or not out. It is false to reason that a batsman bats better when he works on (what will become) a not-out score.

For each batsman, each of the three distributions was fitted. The Kolmogorov-Smirnov statistic D has been calculated for every fit after replacing each not-out score by a complete score according to the procedure explained in part 4. A few batsmen had so many zero scores that the first D -value was the largest, irrespective of the distribution fitted. In such cases the second largest D -value was used as test statistic. The p -values have been calculated by means of the approximate formula

$$p = 2.d \sum_{j=0}^{[n(1-d)]} \binom{n}{j} \left(1 - d - \frac{j}{n}\right)^{n-j} \left(d + \frac{j}{n}\right)^{j-1}$$

where d is the observed value of the test statistic and $[x(1-d)]$ the greatest integer less than or equal to $x(1-d)$, given in Conover (1999) p. 432. It was found that for the great majority of batsmen the gamma distribution gave the best fit. At the 5% level of significance 18 of the batsmen had good gamma and Weibull fits with the gamma the best of the two in 14 cases. The exponential distribution gave poor fits in 6 of the 18 cases (see the p -values in tables 2 and 3). Some of these batsmen, who had very good fits, had to be excluded from the further study (Smith, Gayle, Trescothick) because they had very few not-out scores, which created a problem regarding their downscale variables in part 4. The top nine batsmen (down to Lara) with the best fits according to the gamma model, and who had all played more than 160 innings, were used for the further study. The methods used in this study are critically dependent on very good distributional fits and therefore the study is limited to these nine batsmen. The gamma and Weibull fits of the top nine batsmen are all very good. No further use will be made of exponential fits.

4. Finding the best estimator for the mean

For each not-out score x_i^* the expected complete score X_i^* can be found by using the formula $X_i^* = E(X \mid X \geq x_i^*)$ – see Lemmer (2001) for the formulae. This is possible because the parameters of the distributions (gamma and Weibull) of the scores of the batsman are known. The estimated average is

$$\bar{x}_a = \left(\sum_{i=1}^m x_i + \sum_{i=m+1}^n X_i^* \right) / n.$$

The true average is obtained by using the parameters of the distribution, i.e. $\bar{x}_b = \alpha/\lambda$ for the gamma distribution and $\bar{x}_b = \Gamma\left(1 + \frac{1}{\beta}\right)/\lambda$ for the Weibull distribution. The values of the parameters, the p-values, ave, $\bar{x}_a$, $\bar{x}_b$ and mean(u) – to be explained below – are given in tables 2 and 3.

The method used to find the X^* values seems to work well because in most cases the values of $\bar{x}_a$ and $\bar{x}_b$ are very similar. In the top parts of tables 2 and 3 ave generally has values close to the true ($\bar{x}_b$) values for the nine players with very good fits because most of these batsmen have small proportions of not-out scores. It will, however, be shown later in this study, that ave becomes very unreliable when propno becomes large.

In order to develop an estimator for the mean *if the distribution is unknown*, calculate the downscale variable $u_i = x_i^*/X_i^*$ for $i = m + 1, \dots, n$ and call mean(u) the (average) downscaling value for the batsman. Based on the assumption made before, mean(u) can be expected to have a value close to 0.5, but as can be seen in tables 2 and 3, they vary between batsmen. A very simple argument is now used. If mean(u) is the average downscaling from the complete score X^* to the not-out score x^* , its inverse $f = 1/\text{mean}(u)$ can be used to estimate X^* by scaling x^* up if the latter is known.

The value $f = 2$ mentioned before thus relates to the theoretical case where the downscale random variable U has a distribution with $E(U) = \frac{1}{2}$. A more general method to get an estimator for the mean is to estimate X_i^* from x_i^* by using available statistics.

Table 2. Parameter values and estimates of the mean based on the gamma distribution in ODIs

Name	α	λ	p-value	ave	$\bar{x}_a$	$\bar{x}_b$	mean(u)
Cairns	1.02759	0.03500	0.746	29.43	29.37	29.36	0.4337
Kallis	0.81008	0.01675	0.735	45.41	47.10	48.36	0.4740
Atapattu	0.80598	0.02077	0.712	37.74	38.52	38.81	0.5133
Gibbs	0.67080	0.01841	0.683	35.53	36.44	36.44	0.5173
Fleming	0.89390	0.02757	0.586	32.12	32.36	32.42	0.5969
Dravid	1.00845	0.02359	0.581	40.19	40.48	42.75	0.4800
Chanderpaul	1.06530	0.02940	0.512	36.42	36.23	36.23	0.4966
Razzaq	0.82665	0.02641	0.482	30.25	31.30	31.30	0.3967
Lara	0.82825	0.01932	0.447	41.95	42.58	42.87	0.4896
Gayle	0.69082	0.01753	0.379	38.74	39.40	38.39	0.6594
Smith	1.17105	0.02886	0.378	40.77	40.57	39.46	0.6203
Trescothick	0.67001	0.01742	0.281	37.73	38.46	38.44	0.5844
Martyn	0.73069	0.01596	0.211	42.24	45.78	45.75	0.3676
Kaluwitarana	0.70825	0.03132	0.210	22.22	22.61	22.61	0.4032
Boucher	0.77318	0.03028	0.185	24.47	25.53	25.45	0.3432
Vettori	0.78112	0.05373	0.161	13.69	14.54	14.54	0.3225
Streak	0.87742	0.02999	0.135	28.20	29.26	29.25	0.3744
Klusener	0.73907	0.01625	0.106	41.10	45.48	45.45	0.3566
Lehmann	0.94479	0.02406	0.043	38.96	39.27	39.25	0.3743
Harris	0.84881	0.02814	0.038	29.00	30.16	30.15	0.4211
Ponting	0.84718	0.01994	0.020	41.86	42.48	42.47	0.4012
Pollock	0.80949	0.03093	0.008	24.67	26.17	25.79	0.3096

For each batsman there is a strong relationship between x_i^* and X_i^* , e.g. $X^* = 51.91 + 1.047x^*$ for Kallis in his gamma model (similar to $X^* = E(X) + x^*$ in the case of the exponential distribution). By using the (x^*, X^*) data of all nine batsmen jointly and also their statistics such as avno, no, avout, out and propno, it is possible to find various estimators for the mean depending on the structure used and which statistics are included in

the regression model. Among the estimators considered were ave, KH (the Kimber and Hansford (1993) estimator), $e_2 = (\sum_{i=1}^m x_i + 2 \cdot \sum_{i=m+1}^n x_i^*)/n$, which can be written as $e_2 = (\text{sumout} + 2 \cdot \text{sumno})/n$, $e_4 = (\text{sumout} + 1.025 \text{sumno} + f_4 \cdot \text{no})/n$ where $f_4 = 0.6 \text{avout} + 0.3 \text{avno} + 54 \text{propno}$, $e_5 = (\text{sumout} + f_5 \cdot \text{sumno})/n$ where $f_5 = 1.8 - 0.005 \text{avno} + 0.4 \text{propno}$ and $e_6 = (\text{sumout} + f_6 \cdot \text{sumno})/n$ where $f_6 = 2.2 - 0.01 \text{avno}$. It was found that estimators having the structure $(\text{sumout} + f \cdot \text{sumno})/n$ were much better than others such as e_4 .

Table 3. Parameter values and estimates of the mean based on the Weibull distribution in ODIs

Name	β	λ	p-value	ave	$\bar{x}_a$	$\bar{x}_b$	mean(u)
Cairns	1.02959	0.03373	0.804	29.43	29.30	29.30	0.4377
Kallis	0.87022	0.02182	0.646	45.41	47.72	49.15	0.4599
Atapattu	0.86900	0.02734	0.598	37.74	38.87	39.26	0.4989
Gibbs	0.75476	0.03185	0.585	35.53	37.08	37.19	0.4808
Fleming	0.93069	0.03176	0.532	32.12	32.47	32.55	0.5890
Dravid	1.02199	0.02328	0.592	40.19	40.35	42.57	0.4855
Chanderpaul	1.05245	0.02715	0.536	36.42	36.11	36.09	0.5028
Razzaq	0.86724	0.03372	0.410	30.25	31.89	31.87	0.3820
Lara	0.89032	0.02451	0.367	41.95	42.79	43.19	0.4805
Gayle	0.77614	0.02888	0.478	38.74	39.95	40.13	0.6183
Smith	1.15374	0.02361	0.475	40.77	40.40	40.28	0.6437
Trescothick	0.76149	0.02986	0.216	37.73	39.01	39.39	0.5449
Martyn	0.79561	0.02374	0.163	42.24	47.81	47.91	0.3432
Kaluwitarana	0.77955	0.05045	0.205	22.22	22.88	22.89	0.3805
Boucher	0.82946	0.04236	0.120	24.47	26.07	26.09	0.3276
Vettori	0.80482	0.07478	0.109	13.69	15.19	15.09	0.2987
Streak	0.89316	0.03524	0.093	28.20	30.03	29.98	0.3598
Klusener	0.79465	0.02365	0.055	41.10	48.08	48.14	0.3297
Lehmann	0.99395	0.02569	0.055	38.96	39.02	39.03	0.3803
Harris	0.88685	0.03450	0.037	29.00	30.75	30.75	0.4077
Ponting	0.91909	0.02439	0.021	41.86	42.52	42.66	0.3998
Pollock	0.85315	0.04033	0.005	24.67	26.90	26.91	0.2961

In order to study the behaviour of estimators in the case of large propno values (for which no real cases are available) some of the out scores will be scaled down to become not-out scores. An important question is how to scale out scores down to supposedly not-out scores. If the x^* are estimated from the X^* in the data set, the relationship is $x^* = -49.78 + 0.956X^*$ in Kallis' case. This is not useful because for small or moderate X^* values the estimates of x^* are negative. More importantly, **the relationship between X^* and x^* will not remain the same if propno changes**. Another plan has to be made. For each batsman the set of downscale values $u_i = x_i^*/X_i^*$ have been computed. Those of Kallis (who normally bats at 3 or 4) varied between 0.04 and 0.69 with average 0.474, whereas Razzaq's (who bats at 8 or 9) u_i 's vary between 0 and 0.75 with average 0.3967. The u -values depend on the characteristics of the batsman, which includes his batting position. To model the situation where the batsman scores a high percentage of not-out scores in a short series (e.g. Klusener in the 1999 World Cup Series), these u -values are the appropriate ones to use. Unfortunately it is not possible to obtain reliable results (e.g. to fit a distribution to the scores) by modelling a small number of innings played. It is necessary to consider the situation where all the batsman's scores are used, including a proportion of his out scores that have to be scaled down by using the u -values in order to get the desired propno value. **Conceptually** the rules of the game have to be changed to model the situation where the batsman obtains a large proportion of not-out scores. The procedure is to determine how many of the batsman's out scores should be scaled down to not-out scores in order to get a specific percentage of not-out scores, say propno = 0.4. For Razzaq 67 not-out scores are required, so $67 - 41 = 26$ out scores have to be scaled down. Draw a random sample (without replacement) of 26 out scores and a random sample of 26 u -values (with replacement), multiply each out score by

a u -value to get the additional not-out scores. Use his full transformed data set to calculate the values of all the estimators and the necessary statistics for comparison purposes. The same procedure is repeated for every batsman. The selection process is as follows.

Let $\hat{\theta}$ be an estimator of the parameter θ . $\hat{\theta}$ is unbiased if $E(\hat{\theta}) = \theta$. The most logical way to compare different estimators of θ is to calculate the mean squared error

$$\begin{aligned} MSE(\hat{\theta}) &= E(\hat{\theta} - \theta)^2 = var(\hat{\theta}) + (E(\hat{\theta}) - \theta)^2 \\ &= var(\hat{\theta}) + B^2 \text{ where } B = E(\hat{\theta}) - \theta \end{aligned}$$

is the bias of the estimator. In an extensive study a large number of possible estimators have been compared by calculating their averages and standard deviations based on 1000 repetitions, the values of the MSE and B of each estimator for all the batsmen for the gamma and Weibull distributions. The final choice, however, is made by using a standardized MSE . The reason for standardization is that if two players have similar MSE 's but their true averages differ (e.g. Kallis with $\bar{x}_b = 48.36$ and Cairns with $\bar{x}_b = 29.36$) it is better to work with $\sqrt{MSE}/\bar{x}_b$ or more conveniently with $RMS = 100\sqrt{MSE}/\bar{x}_b$. The average RMS value for all nine batsmen is calculated for each estimator for each propno value. In the final selection thirteen estimators have been studied. The results of the best three and also those of ave and KH , are given in table 4. They are ranked according to the average RMS values in the case of the gamma distribution. According to the minimum RMS criterion, the estimator e_6 is the best with the very simple estimator e_2 second, but ave and KH are very bad. Although the final choice of the best estimator has been made according to RMS values, it is also interesting to look at the bias of the estimators. The bias B is also standardized by using

$SB = 100.B/\bar{x}_b$. Also according to the average SB of the estimators, which is also given in table 4, e_6 and e_2 are very good.

Table 4. Average RMS and SB values of estimators for the gamma and Weibull fits

Rank	propno	Estimator	Ave RMS Gamma	Ave RMS Weibull	Ave SB Gamma	Ave SB Weibull
1	0.40	e_6	4.52	4.59	-0.05	-1.08
2	0.40	e_2	4.71	4.44	1.68	0.57
3	0.40	e_5	5.16	5.86	-3.06	-4.07
4	0.40	KH	20.45	19.33	20.34	19.22
5	0.40	ave	23.36	22.18	23.24	22.05
1	0.50	e_6	5.09	5.04	0.31	-0.83
2	0.50	e_2	5.36	5.02	1.52	0.30
3	0.50	e_5	5.43	5.96	-2.23	-3.36
4	0.50	KH	34.45	33.09	34.33	32.97
5	0.50	ave	40.29	38.84	40.17	38.72
1	0.80	e_6	6.81	6.61	1.00	-0.60
2	0.80	e_5	7.07	6.65	1.87	0.21
3	0.80	e_2	7.20	6.89	1.12	-0.57
4	0.80	KH	141.67	139.02	139.54	136.96
5	0.80	ave	188.43	183.86	188.26	183.69

Ave and KH , on the other hand, become totally unreliable when propno increases. Similar results have been obtained in the case of the Weibull distribution with e_2 in some cases better than e_6 . Both e_2 and e_6 are very reliable and easy to calculate, and either should be used instead of ave. The overall recommendation is to use the slightly more complicated e_6 in serious studies, but for everyday use e_2 is good enough.

In the initial study of this problem a large set of complete scores were generated from each batsman's distribution and the required proportion scaled down to become not-out scores. The relative performances of the estimators were the same as in this study.

Other methods of downscaling have also been examined. Under the assumption that a batsman's innings has a 0.5 probability of being curtailed

at any stage, it can be conjectured that the downscale random variable, U , has a uniform $[0,1]$ distribution. By using such downscale values, very similar results have been obtained as those given above. This was also the case with three more downscale methods.

5. Conclusions regarding ODI estimators

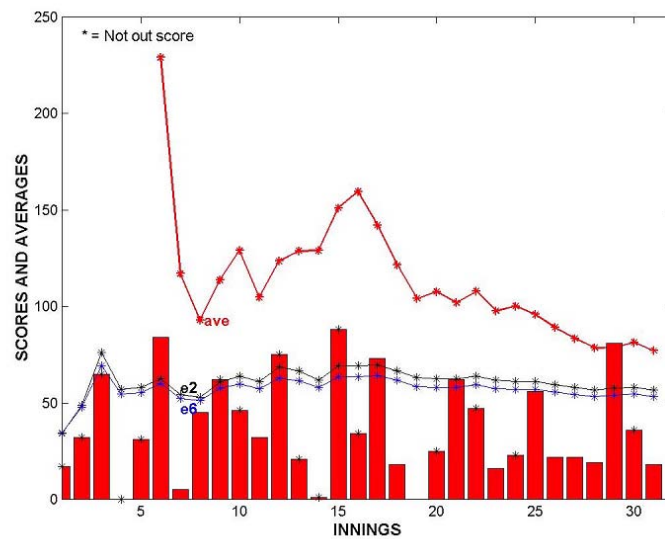
In a test or ODI series the batsmen have batted in a small number of innings. It then often happens that a batsman was not out in a large percentage of his innings. The estimators e_2 and e_6 are both reliable estimators of the average, but ave definitely not. Returning to Lance Klusener's scores of 12*, 52*, 48*, 52*, 46*, 4, 36 and 31* in the 1999 World Cup Series, it is found that his traditional ave = 140.5, but $e_6 = 59.17$ and $e_2 = 65.25$ are much more realistic estimates. There is absolutely no doubt that ave should not be used if propno is large.

A further method to show how erratic ave behaves, is to calculate its value after each innings for the very spectacular Australian batsman Michael Hussey, using his ODI scores, starting from the sixth (his first five scores were all not out). From table 5 and figure 1 (where an asterisk on a bar indicates a not-out score) it is clear that e_2 and e_6 are much more stable than ave, which still jumps around even up to the 31st score.

By examining similar plots for other batsmen it was confirmed that ave moves closer to the other two estimates as the number of innings played increases, but the closeness depends on the value of propno. Even after many innings played, a sequence of not-out scores causes the ave curve to move away from the other two.

Table 5. Comparison between estimators of the average for M. Hussey in ODIs

Innings	6	7	8	9	10	11	12	13	14
Ave	229.0	117.0	93.0	113.7	129.0	104.8	123.5	128.8	129.0
e_2	62.3	54.1	53.0	60.9	64.0	61.1	68.5	66.5	61.9
e_6	60.2	52.3	51.4	57.6	59.9	57.4	62.8	61.4	58.1
Propno	0.83	0.71	0.63	0.67	0.70	0.64	0.67	0.69	0.71
Innings	15	16	17	18	19	20	21	22	23
Ave	151.0	159.5	142.2	121.5	104.1	107.7	102.0	107.9	97.7
e_2	69.5	69.4	69.6	66.7	63.2	62.6	62.5	64.0	61.9
e_6	63.7	63.7	64.2	61.7	58.4	58.0	58.2	59.3	57.4
Propno	0.73	0.75	0.71	0.67	0.63	0.65	0.62	0.64	0.61
Innings	24	25	26	27	28	29	30	31	
Ave	100.2	95.8	89.1	83.5	78.5	78.7	81.3	77.1	
e_2	61.2	61.0	59.5	58.1	56.7	57.6	58.0	56.7	
e_6	57.0	57.0	55.6	54.4	53.1	54.1	54.5	53.3	
Propno	0.63	0.60	0.58	0.56	0.54	0.52	0.53	0.52	

**Figure 2.** Different estimates of the ODI average until 06/04/2006: M. Hussey

The desire to see whether real career data confirm the findings of this study is hampered by the fact that very few batsmen have high proportions of not-out scores. In the initial data set the highest value of propno was 0.37. In order to see how the estimators perform, some results are given in table 6, but these cannot be used to judge the merits of the estimators. The yardstick $\bar{x}_b$ used to compare the estimates with, is taken from each batsman's best fit distribution. It is crucial that $\bar{x}_b$ should be reliable, therefore the players who do not have good enough fits (from Boucher to Pollock) are excluded. In table 6 the estimate closest to $\bar{x}_b$ is indicated by an asterisk. For small propno values ave remains good (in terms of bias), but if propno becomes much larger, it has been shown that ave should no longer be used. Both e_2 and e_6 are simple to calculate and any cricket enthusiast should be able to calculate them. The very simple e_2 is strongly recommended for general use.

Table 6. Comparison of estimators in ODIs

Name	n	propno	ave	e_2	e_6	$\bar{x}_b$
Cairns	184	0.120	29.43	29.61	29.20*	29.30
Kallis	206	0.180	45.41	48.04*	43.72	48.36
Atapattu	223	0.117	37.74*	40.42	37.54	38.81
Gibbs	168	0.077	35.53	37.77	35.55*	36.44
Fleming	226	0.084	32.12	35.01	32.41*	32.42
Dravid	236	0.123	40.19	41.53*	39.58	42.57
Chanderpaul	166	0.127	36.42	37.68	36.13*	36.09
Razzaq	168	0.244	30.25	30.46*	29.62	31.30
Lara	249	0.104	41.95	43.60*	41.32	42.87
Gayle	120	0.050	38.74*	42.35	37.29	40.13
Smith	67	0.045	40.77	42.00	40.52*	40.28
Trescothick	108	0.056	37.73*	40.48	37.21	38.44
Martyn	157	0.306	42.24	42.27*	39.38	45.75
Kaluwitharan	181	0.077	22.22	23.11	22.75*	22.61

The estimators e_2 and e_6 have disadvantages in the sense that a small not-out score can cause the estimators to decrease, and a large not-out score can cause a large increase in their values. They are nevertheless much more satisfactory than ave .

6. Test matches

In the case of test matches the same procedure as for ODIs has been followed, but much less detail need be given. The test match scores of twenty experienced batsmen who have been drawn randomly from all current test players who had played at least one hundred innings each and had averages above twenty, have been taken from Cricinfo (2005b), also on 18/08/2005. The statistics of these players are given in table 7.

Table 7. Statistics of test data set

Name	n	avout	out	avno	no	propno	ave
Kallis	155	40.30	129	82.23	26	0.168	56.88
Langer	156	41.54	148	82.50	8	0.051	46.00
Inzamam	168	42.08	150	72.67	18	0.107	50.80
Trescothick	119	42.61	109	29.00	10	0.084	45.27
Dravid	151	48.71	133	67.67	18	0.119	57.86
Ganguly	133	35.15	121	58.00	12	0.090	40.90
Ponting	149	47.35	129	55.85	20	0.134	56.01
Fleming	164	33.14	154	88.20	10	0.061	38.86
Jayawardene	113	43.71	104	45.44	9	0.080	47.64
Hussain	171	32.39	155	46.50	16	0.094	37.19
Cairns	104	32.29	99	24.60	5	0.048	33.54
Hayden	123	48.45	113	39.30	10	0.081	51.93
Gibbs	117	44.02	111	56.83	6	0.051	47.09
Khan	104	25.29	96	39.13	8	0.077	28.55
Laxman	103	35.90	92	59.82	11	0.107	43.05
Thorpe	179	34.54	151	54.57	28	0.156	44.66
Vaas	124	16.02	102	24.00	22	0.177	21.20
Boucher	113	26.62	98	26.53	15	0.133	30.68
Vaughan	105	38.23	97	50.38	8	0.076	42.38
Jacobs	112	17.59	91	46.48	21	0.188	28.32

The three distributions were fitted to the data. It was again found that the gamma distribution gave the best fit and the exponential distribution by far the worst. The parameter values, p -values, the averages according to the fitted distribution ($\bar{x}_b$) and also after augmenting not-out scores ($\bar{x}_a$), and also the traditional averages, were calculated. The nine batsmen with the best fits and who had played at least 130 innings each, were used for the extensive process of finding the best estimators. Their details are given in tables 8 and 9.

Table 8. Parameter values and estimates of the mean based on the gamma distribution in tests

Name	α	λ	p -value	ave	$\bar{x}_a$	$\bar{x}_b$	mean(u)
Kallis	0.72843	0.01226	1.000	56.88	59.41	59.42	0.4747
Langer	0.73174	0.01571	0.999	46.00	46.58	46.58	0.5380
Inzamam	0.73776	0.01417	0.885	50.80	52.06	52.06	0.4687
Dravid	0.78683	0.01332	0.769	57.86	59.07	59.07	0.4496
Ganguly	0.74000	0.01773	0.733	40.90	41.74	41.74	0.4819
Ponting	0.79912	0.01399	0.709	56.01	57.12	57.12	0.3976
Fleming	0.69895	0.01767	0.644	38.86	39.56	39.56	0.5471
Hussain	0.72824	0.01922	0.602	37.19	37.88	37.89	0.3888
Thorpe	0.75822	0.01644	0.274	44.66	46.09	46.12	0.4214

Table 9. Parameter values and estimates of the mean based on the Weibull distribution in tests

Name	β	λ	p -value	ave	$\bar{x}_a$	$\bar{x}_b$	mean(u)
Kallis	0.80277	0.01850	0.812	56.88	60.90	61.09	0.4486
Langer	0.80983	0.02391	0.924	46.00	46.91	46.98	0.5141
Inzamam	0.81366	0.02118	0.683	50.80	52.72	52.86	0.4478
Dravid	0.85511	0.01817	0.663	57.86	59.53	59.65	0.4369
Ganguly	0.82045	0.02629	0.721	40.90	42.14	42.35	0.4623
Ponting	0.85871	0.01877	0.566	56.01	57.58	57.60	0.3862
Fleming	0.78289	0.02866	0.547	38.86	40.05	40.16	0.5166
Hussain	0.80788	0.02930	0.446	37.19	38.25	38.40	0.3716
Thorpe	0.83127	0.02353	0.221	44.66	46.74	46.90	0.4054

Note that the proportion of not-out scores is much lower than in the case of ODIs – the highest is 16.8%. It was again necessary to obtain additional not-out scores, and this has been done in exactly the same manner as before. Based on extensive preliminary studies, the best estimators were

$$e_2 = \left(\sum_{i=1}^m x_i + 2 \sum_{i=m+1}^n x_i^* \right) / n,$$

$e_5 = (\text{sumout} + f_5 \text{ sumno})/n$ with $f_5 = 2.25 - 0.012\text{avno} + 0.10\text{propno}$ (this is different from f_5 for ODIs) and $e_8 = (\text{sumout} + f_8 \text{ sumno})/n$ with $f_8 = 2.2 - 0.01\text{avno} + 0.15\text{propno}$.

In a study similar to that in the ODI case, the best estimators among the final set of eleven estimators have been determined by calculating standardized mean squared errors and biases. The average *RMS* and *SB* values of *ave*, e_2 , e_5 , e_8 and *KH* for the nine batsmen are given in table 10.

Table 10. Average *RMS* and *SB* values of estimators for gamma and Weibull fits

Rank	propno	Estimator	Ave RMS Gamma	Ave RMS Weibull	Ave SB Gamma	Ave SB Weibull
1	0.10	e_2	4.40	4.55	0.03	-2.36
2	0.10	e_8	4.70	5.46	-5.81	-3.70
3	0.10	e_5	5.10	5.94	-3.04	-4.51
4	0.10	<i>ave</i>	23.93	21.40	23.71	21.15
5	0.10	<i>KH</i>	26.73	24.10	26.51	23.86
1	0.50	e_8	5.28	5.96	-5.24	-3.58
2	0.50	e_2	5.28	5.62	-0.56	-3.27
3	0.50	e_5	5.62	6.44	-2.05	-4.49
4	0.50	<i>ave</i>	39.88	36.63	39.65	36.38
5	0.50	<i>KH</i>	41.84	38.51	41.56	38.22
1	0.80	e_8	7.25	7.50	0.45	-2.99
2	0.80	e_5	7.35	8.07	-1.17	-4.50
3	0.80	e_2	7.84	8.68	-2.36	-5.95
4	0.80	<i>KH</i>	151.96	143.17	149.81	141.23
5	0.80	<i>ave</i>	181.73	172.27	181.41	171.97

In the case of test matches e_8 is recommended for serious studies, but the simpler e_2 for general everyday use. For small propno values ave remains good (in terms of bias), but if propno becomes much larger than, say, 0.20, ave should no longer be used.

7. Measure of batting performance in a series of ODIs or test matches

A batsman's batting performance depends not only on the average he achieves, but also on his consistency and strike rate – cf. Lemmer (2004) for a discussion of the importance and use of the consistency coefficient and strike rate in defining a batting performance measure. In the case of a small number of innings it is necessary to adjust the way these quantities are used to define a measure of batting performance, BP^* . The procedure is to calculate the consistency coefficient CC and strike rate SR before the start of the series. Then add the series scores and calculate these quantities once more, indicated by CC^* and SR^* . Due to the fact that the consistency coefficient reacts quickly to changes in performance, the ratio $C = CC^*/CC$ will reflect any changes in consistency during the series. Similarly, calculate $R = SR^*/SR$ to measure the change in strike rate. As in Lemmer (2004), let $RP = R^{0.43}$ and define the measure of batting performance in the case of ODIs by $BP^* = e_6.C.RP$. Here the batsman's consistency and strike rate just before the start of the series is used as yardstick to measure changes in these characteristics during the series. After completion of an ODI series, BP^* should be used to compare the batting performances of the batsmen rather than the very naive traditional average.

Before the series of five ODIs against South Africa during February and March 2006, Michael Hussey had an average of 100.22, $e_6 = 57.00$, a

strike rate of 99.78 and a consistency coefficient $CC = 2.3143$. He did not perform equally spectacular in the series and had the following career figures after the series: Average = 78.71, $e_6 = 54.07$, $SR^* = 99.27$ and $CC^* = 2.2793$. These gave $C = 2.2793/2.3143 = 0.9849$, $R = 99.27/99.78 = 0.9949$, $RP = (0.9949)^{0.43} = 0.9978$. Both his consistency and strike rate went down slightly, although both were still very good. For the series he obtained $e_6 = 40$ and his batting performance measure was $BP^* = 40 \times 0.9849 \times 0.9978 = 39.31$.

For test matches, the estimated average e_8 is adjusted by using the consistency and strike rate update measures $C = CC^*/CC$ and $R = SR^*/SR$ immediately after conclusion of the series. The measure of batting performance in the series is then defined by $BP^* = e_8.C.RP$ where $RP = R^{0.30}$ as in Lemmer (2004).

8. Conclusion

The finding in this study that the traditional average should be replaced by an equally simple but much more stable and statistically sound estimator of the average in the case of moderate and high proportions of not-out scores, is very important. Cricket statisticians, who know nothing about variability, will reason that for the majority of applications the values of ave, e_6 (or e_8) and e_2 are very similar. The best hope of convincing them to use e_2 would probably be to refer to table 5 and figure 2, and show them how a single additional not-out score can cause ave to increase drastically, e.g. from 129 to 151 (innings 14 to 15), whereas e_2 only changes from 61.9 to 69.5 and e_6 from 58.1 to 63.7.

The results of this study can be summarized in the following recommendation. Use e_6 in the case of limited overs matches and e_8 for

unlimited overs matches. For easy and fast results, e_2 may instead be used without much loss of accuracy.

When a batsman is selected for his international team, his first series of scores are crucial, because they determine his fate of being kept or dropped. It is therefore essential to use the best available performance measure when players' performances are compared after a short series of matches. The notoriously bad behaviour of ave in the case of not-out scores rules it out as a suitable measure.

Acknowledgements

The author is grateful for the valuable comments of the referee and co-editor that led to a substantial improvement of the paper.

References

- BARR, G.D.I. & VAN DEN HONERT, R. (1998). Evaluating batsmen's scores in test cricket. *South African Statist. J.*, **32**(2), 169–183.
- BUCKLEY, J. & JAMES, I. (1979). Linear regression with censored data. *Biometrika*, **66**(3), 429–436.
- CONOVER, W.J. (1999). *Practical nonparametric statistics*. 3rd edition. John Wiley & Sons, New York.
- CRICINFO (2005a). ODI career batting averages. [http://www.cricinfo.com/link_to_database/STATS/]. Retrieved 18 August 2005.
- CRICINFO (2005b). Test career batting averages. [http://www.cricinfo.com/link_to_database/STATS/]. Retrieved 18 August 2005.
- KIMBER, A.C. & HANSFORD, A.R. (1993). A statistical analysis of batting in cricket. *J.R. Statist. Soc. A*, **156**(3), 443–455.

- LEMMER, H.H. (2001). Beraming van die kolfvermoë van Suid-Afrikaanse internasionale eendagkrieketspelers. *Suid-Afrikaanse Tydskrif vir Natuurwetenskap en Tegnologie*, **20**(2), 45–48.
- LEMMER, H.H. (2004). A measure for the batting performance of cricket players. *SA Journal for Research in Sport, Physical Education and Recreation*, **26**(1), 55–64.